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To cite this article: Ji Yang Lee & Cheol-soo Lee (2018) Path planning for SCARA robot based on marker detection using feature extraction and, labelling, International Journal of Computer Integrated Manufacturing, 31:8, 769-776, DOI: [10.1080/0951192X.2018.1429669](https://doi.org/10.1080/0951192X.2018.1429669)

To link to this article: <https://doi.org/10.1080/0951192X.2018.1429669>



Published online: 24 Jan 2018.



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ARTICLE



Path planning for SCARA robot based on marker detection using feature extraction and, labelling

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ABSTRACT

This article addresses selective compliance assembly robot arm (SCARA) robot path planning using markers. Path planning requires the positions and orientations of goal points. A path planning method using markers finds the position and orientation of the markers and recognises the markers one by one to get to the goal points. This method uses two image processes: which are Speeded-Up Robust Features (SURF) and labelling. To obtain the 3D positions of the markers, the coordinate space is transformed from colour space to camera space using a Kinect sensor. This process was implemented on a SCARA robot system. By using D-H (Denavit–Hartenberg) parameters, the SCARA robot's inverse kinematics transforms the extracted points into joint actuator trajectories. After finding the position and orientation of the markers, the path planning is performed. The time-efficiency of the proposed method is validated through experimental results.

ARTICLE HISTORY

Received 28 November 2016
Accepted 3 January 2018

KEYWORDS

Path planning; feature extraction; computer vision; robot application; SCARA robot; Kinect sensor

1. Introduction

Industrial robots have become important role in many fields. Their importance are rooted in flexibility (Kaltsoukalas, Makris, and Chryssolouris 2015). It is possible that industrial robots' movements are free and their roles are changeable as what they hold as end-effect. On the other hand, robots are highly efficient with repeatability and durability because, they can alternate with people's role and movements. Therefore, industrial robot is adopted in broad fields and the demands of robots have been followed.

However, there are some problems with applying and installing. The most frequently used path planning method is the lead through method, which is time-consuming and usually requires experienced and qualified experts with teaching pendent (Lin et al. 2015). Usually, it is efficient way to operate industrial manipulators. However, it takes a lot of money and time. Furthermore, it impedes the popularisation of robotic system on various fields. Also, there is no flexibility with this path planning method. It is a contradiction that the ultimately flexible tool doesn't have flexibility. In other words, easy robotic path planning is required.

There are many ways to solve the path planning problem (Tsarouchi, Makris, and Chryssolouris 2016). First of all, the computed model is convenient way to make path planning because, except external factors, it is robust method to figure out system. Erdős et al. (2016) suggested semi-automated path planning for robotic laser welding systems. Chung and Lee (2005) used an augmented Petri net to model and control assembly tasks and compute the condition and costs of state for motion planning. Mei and Lee (2016) proposed an octree-based collision detection method using virtual reality with machine tools and robot arms. All these researches use various

ways to use CAD model for path planning. However, it is impossible to make all the CAD models for any system. Furthermore, the amateurs are not used to using CAD modelling.

Also, there is another way to resolve the problem with using physical data. First of all, physical data, such as position, torque and joint angle, are important for collaborative robots to work with people. 'Baxter' is an example of the collaborative robots. Path planning of this collaborative robot rely on dragging it by worker (Maurtua et al. 2017). For this action, physical data, torque, angle etc. are absolutely necessary. Also, using physical data need real-time digital signal processing. Moreover, workers need to learn about using specific system and it is inevitable to touch robot.

Vision system is well-used tool when avoiding contacts with robot for hazardous and cleanliness-sustained environment (Heyer 2010; Luk et al. 2005; Oh et al. 2009). Vision system is non-contact sensor system and it is fast to figure out what it senses. Moreover, sensor's cost and quality have been improved, driven by technological shifts. It has been chosen more frequently for automated system. Especially, the depth value for each pixel scale is known with using two vision sensors. This value can be used for configuring not for two dimensions but for three dimensions. These spatial information help path planning using vision sensors (Zhang, Wang, and Chan 2015; Sridhar, Oulasvirta, and Theobalt 2013).

For instance, Zheng et al. (2013) used a vision sensor to find the position and orientation of a mass product from an image. Yuan et al. (2015) introduced the idea of automatic robot taping, where a Kinect sensor was used as a 3D scanner to acquire 3D point clouds of target objects for taping; the data were then used to tape using a 6-DOF robot arm. Park et al. (2015) proposed a parallel Cartesian planning algorithm that

precomputes a roadmap to take into account static obstacles in the environment, as well as singular configurations. Similar to the aforementioned examples, the vision system is good way to acquire spatial data.

This article suggests the path planning method for selective compliance assembly robot arm (SCARA) robot using marker detection. Robot's path planning is difficult for amateurs and software and equipment for path planning is expensive and tough to use for them. Therefore, developed methods use Microsoft Kinect sensor and simple markers for SCARA robot's path planning.

2. Implementation system

2.1. Implementation system

The setup of the proposed system is shown in Figure 1. SCARA robot is used in this article. The SCARA robot has four joints – three rotational and one linear joints – and it is usually used to transport lightweight objects (Das and Dülger 2005; Isioli and Legnani 2002). It includes also a pneumatic gripper, a robot controller, a desktop, the Kinect sensor and markers.

2.2. Marker

The marker is a tool used frequently to build augmented reality. In an image, a marker gives detailed information about the space (Cherubini et al. 2013). In this article, markers are used as a tool to transfer information about the path's points. Every point has 6 degrees of freedom. The primary characteristic of markers is their asymmetry. Asymmetry is an important character for recognising the orientation of the markers in this path planning method. To make a marker asymmetric, it can be composed of a black boundary with a white background and a simple asymmetric image. Examples are shown in Figure 2.

2.3. Kinect sensor

Microsoft's 'Kinect sensor' is a vision sensor. The Kinect sensor can process diverse data and so is utilised in various ways in robotics. The provided images are colour image, depth image,



Figure 2. Example of marker.

infrared (IR) image and body frame image. Object recognition is possible through use of the depth data. In this article, the sensor changes the colour pixel position of the markers into the camera space.

3. Extraction of position and orientation data from markers

3.1. Flow chart

For extraction of position and orientation, the RGB image and the marker images are required. The flow chart for extraction position and orientation data from the markers are composed of four sequences, as shown in Figure 3.

First, the computer obtains the source image, which has the marker images with scenes from the Kinect sensor. Second, two image processes are used: labelling and feature matching. The objective of labelling is to extract markers from the scene. By labelling twice, the markers can be extracted. Through extraction, the position of the markers in the colour image can be processed. The goal of feature matching is to recognise the markers. Each marker has a specific inner image. Therefore, the features in the template can connect to the features in the source image. In this article, feature extraction should be completed before matching. There are some outliers in the feature matching located on the outside of the marker. Third, after removing the outliers, only the connected points in the markers' label remain. Therefore, the label which has the most number of connected points is the marker template. Finally, the markers' positions in the colour image are transposed into the camera space, which is the real coordinate space, through the Kinect SDK. The orientation can

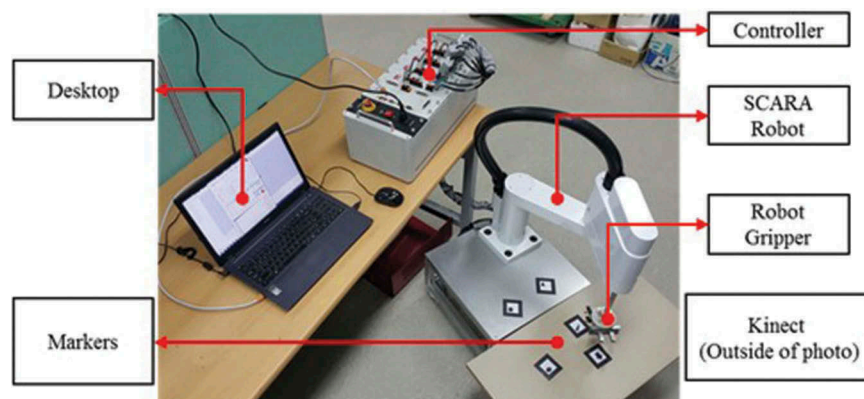


Figure 1. Implementation of H/W system.

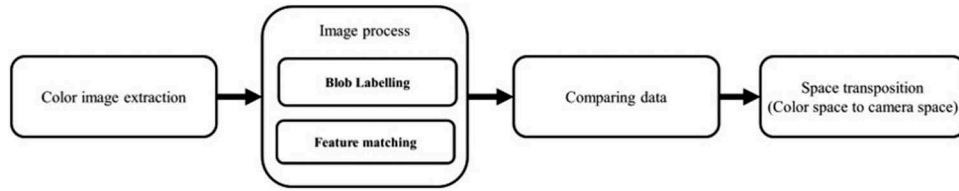


Figure 3. Flow chart for marker position data.

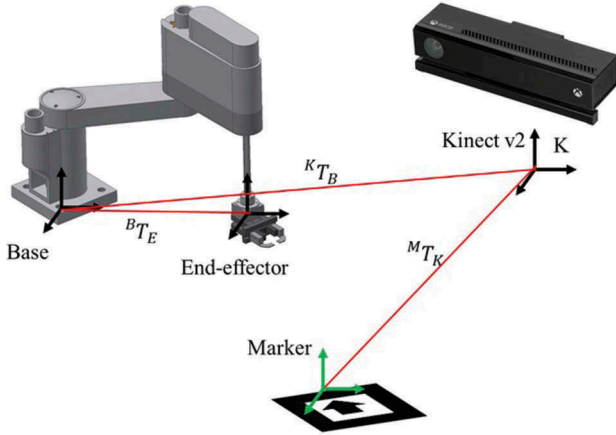


Figure 4. Relationship between different coordinate systems.

easily be calculated through the 4 marker corners because of 3D positioning.

3.2. Image process – blob labelling and feature matching

The extraction process of the markers is composed of blob labelling and feature extraction. Blob labelling is an image process for extracting a blob from a greyscale image. Blob labelling is used twice to extract markers from the source image because of the marker's shape. First, blob labelling extracts the marker outline. Second, it validates markers that have a square in side. For feature matching, feature extraction is a prerequisite. After extracting the features from the source and template images, the features in each image can be connected to the same points.

3.3. Transposition of RGB pixel position to camera space

The pose of the points is defined as Eq. (1)

$$A = [{}^B X_A, {}^B \theta_A] \quad (1)$$

where, ${}^B X_A$ = the position of A based on B and ${}^B \theta_A$ = the orientation of A based on B. The transformation is defined in Eq. (2)

$$C = {}^C T_B A \quad (2)$$

where, ${}^C T_B$ = the transformation matrix, which can change the origin of the pose.

To implement the system, the position of the SCARA robot and the Kinect's base position and orientation are already known. The model of the SCARA robot is already measured

and so are the joint values of the SCARA robot; therefore, the pose of the marker based on the SCARA robot base can be shown with the simple transformation matrix given in Eq. (3), (4), and (5) as shown in Figure 4.

$$K = [{}^K X_M, {}^K \theta_M] \quad (3)$$

$$B = [{}^B X_M, {}^B \theta_M] \quad (4)$$

$$B = {}^K T_B K \quad (5)$$

With these equations, the results of marker detection can be changed to show the end-effector pose.

3.4. Marker position extraction

Before robot path planning, collecting the markers' position is compulsory. The marker extraction was completed through the suggested process in Section 3.1. In Figure 6, an example of marker position extraction is shown. The hexagram-shaped marker is the target in this example. Two markers are detected using blob labelling, represented by the yellow and green lines in the figure. At the same time, feature matching is completed using Speeded-Up Robust Features (SURF) for fast feature extraction (Bay et al. 2008). The feature matching image in Figure 6 shows that the hexagram-shaped marker has 16 counts points in the marker space, which is more than the hexagon-shaped one. By comparing the data phase, with the results of labelling and matched features, the hexagram-shaped marker is identified. Finally, the pixel positions of the marker corners are transposed into real coordinate space through the Kinect SDK. In the example, each point in the colour space is changed into 3-D position data.

4. Path planning using markers

Path planning is primary work for robot to execute jobs. However, it is a complicate job that even needs the experts. The point that makes path planning hard is that the user has to have deep understanding about the goal and the robot. Therefore, this makes user robotic system easy.

The goal of path planning using markers is to make installing system and planning robot's path easy for amateurs. With this path planning, user installs markers as environment in Figure 5. Path planning using markers' flow chart is composed with marker extraction developing in previous section. Figure 5 is flow chart of path planning using markers.

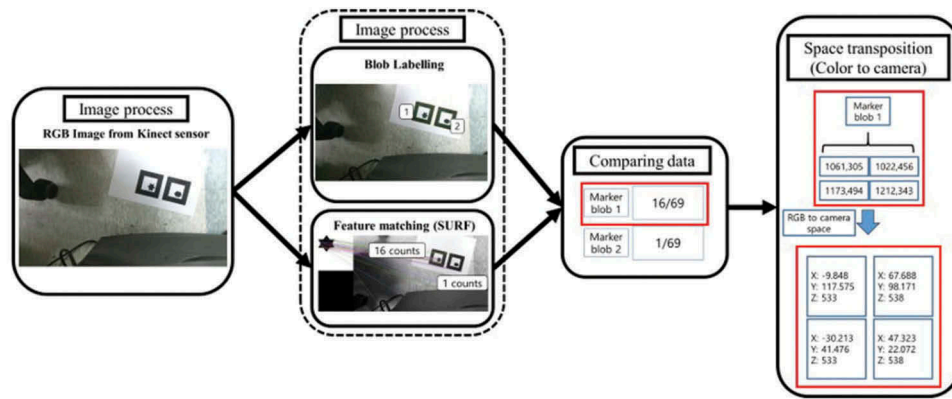


Figure 5. Process of marker detection using markers.

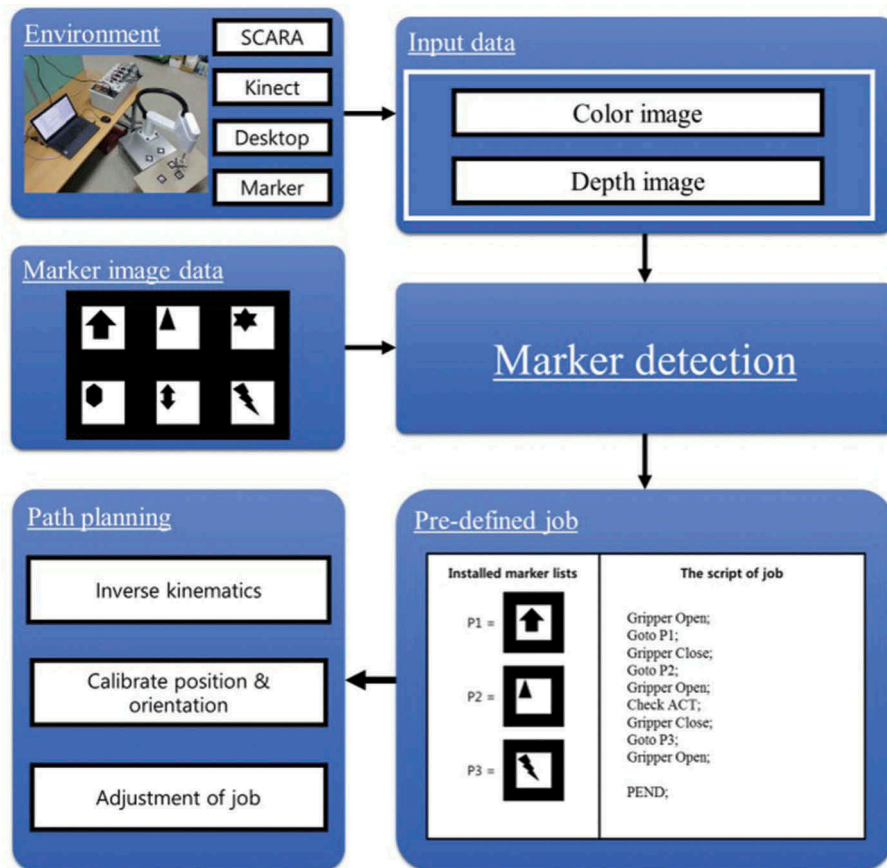


Figure 6. Path planning using markers.

4.1. The preparation of path planning using markers

Path planning with using markers need some preparation. First of all, the hardware components are basic parts, which is introduced in Section 2.1, for marker extraction with robot and other parts for specific implementation system. After installing H/W components, the markers are installed on specific position relative with installed other components.

4.2. Predefined job

Predefined job is the script that is written with relative positions and directions from markers for robot and peripheral devices' operation. Using data from markers after image processing, it is easy to organise the system which works based on the script.

In path planning with using markers, the developer makes predefined job for users because predefined job helps the

users in organising system without knowledge of robotic system. Example of predefined job is shown in Figure 5.

4.3. Marker detection

Two components are required for path planning. One is the image of system on which markers are installed, and the other is the image of markers. Two images are obtained from Kinect sensor. Previously explained in Section 2.3, they are colour image and depth image. Markers should be viewable without truncation in these images

The image of markers is designed with following conditions in Section 2.2. The number of markers are as many as set on the predefined job. The index number for each marker is decided. The index number help sort markers for program and script.

4.4. Path planning using markers

After completion of marker detection, the pose is defined. The objective of path planning is to make a robot carry out a goal. Therefore, a predefined job is needed for path planning using markers. The completed script of the job and the pose of markers can be used to finish the path planning. Finally, the path planning is finished using inverse kinematics. After pre-driving, the operator calibrates the poses and jobs.

5. Implementation – positron emission topography (PET) robotic dispensing system with SCARA robot

PET is a nuclear medicine functional imaging technique (Suetens 2009). This technique observes metabolic processes in the body by detecting gamma rays from a positron-emitting radionuclide (Stauss et al. 2008). It becomes more important that it can explore cancer metastasis with high resolution (Bailey et al. 2005; Analoui, Bronzino, and Peterson 2012).

As becoming more demand of PET, the production system spread much more hospitals. Nuclear medicine experts prepare radiopharmaceuticals for PET with the production system. However, their organs and tissue could be damaged with radioactive exposure even with hot laboratory (Hammer et al. 2013). This is the reason why automation of production system for PET becomes significant (Guillet et al. 2005; Leide-Svegborn 2010).

The production system separate for three levels. One is production of radioactive substance, second step is synthesis of radioactive substance and the last is dispensing process of synthesised product. Especially, dispensing process is the step for dividing highly concentrated synthesised product to appropriate concentration product for each person. Dispensing process is the last step and lower level of radioactive concentration but the closest process to worker. It could make damage to workers.

In this article, Path planning with using marker detection is applied on PET robotic dispensing system with SCARA Robot. Most robotic dispensing system is composed with hot cell at once. If the space is small or the user wants to install the robotic dispensing system to existing hot cell, the appropriate path planning is necessary for each system.

5.1. H/W components in PET robotic dispensing system

PET robotic dispensing system with SCARA robot is composed of various components as shown in Figure 6. Macroscopically, devices are separated in three parts. First, the sCUBE-D is PET dispensing machine. It is the medium to transfer radiopharmaceuticals to small vials from a mother vial, which has the highly concentrated product, with water for dilution.

Second, there are measuring components. Two measuring properties are necessary for precise dispensing. Balance measures weight of vials and dose calibrator measures concentration of radioactivity. According to these two measurements, the system can calculate each vial's weight per concentration of product. Through this calculation, how much of radiopharmaceuticals for the mother vial and each product vial is informationised.

Third, SCARA robot is the robot for moving components, syringe modules to vials. It carries vials for measurement and dispensing. Between these works, the robot moves syringe module for two goals. One is injection of synthesised products and the other is connector of dispensing machine to mother vial.

5.2. Marker detection in PET robotic dispensing system

After making up robotic dispensing system, the 23 markers are installed on the PET robotic dispensing system, as shown in Figure 7. One is initialising position of robot. 15 markers are

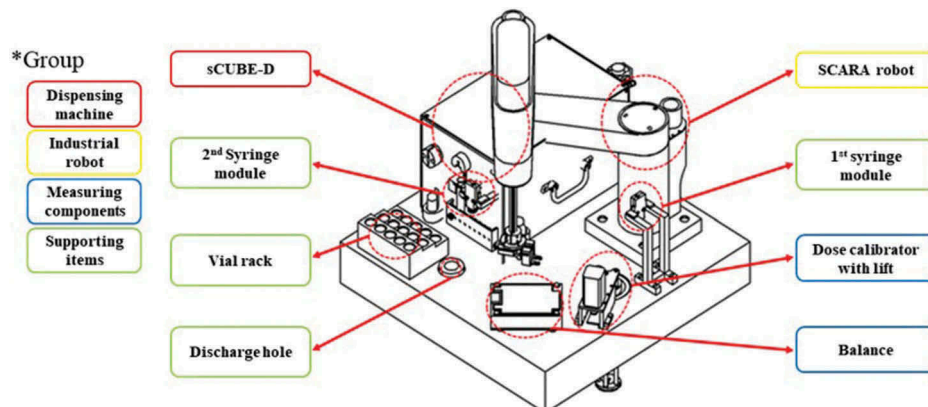


Figure 7. PET robotic dispensing system.

installed on the small vials. Two markers put on balance and dose calibrator. Two Syringe modules have each marker. The others are dispensing position, mother vial storage and discharge hole.

The dispensing process is composed of five phases. First, initialisation phase is preparation of expendables as manually before radiopharmaceuticals supplied. Second, mother vial preparation phase is the step for supplying synthesised product to mother vial. In this phase, the robot move a syringe module for inject product into mother vial and move vial for measurement of weight and activity and, finally, connect mother vial and dispensing machine with the other syringe module. Third, dispensing phase is the step for separating appropriate quantity of product from the mother vial to a product vial. The robot move a product vial to dispensing machine. According to input data and mother vial's calculated data, the product is dispensed to product vials. Fourth, product verification phase is the step for measuring dispensed vial's concentration. It is important phase because of verifying dispensing result and blocking the high-concentrated product to a target patient. Finally, discharging phase is the step for passing the product, which is in the radioactive-free container, to workers. According to each markers and five phases for dispensing process, the predefined job is created.

To take images for marker extraction, Kinect sensor is also fixed for marker extraction. Through images, the markers are extracted and indexed. In Figure 8, markers are extracted and contours are drawn for each marker. The result of extraction is Figure 9.

5.3. SCARA robot path planning using markers

It is impossible to operate robot with just position. To convert the extracted position data into robot joint angles, inverse kinematics should be applied. In this article, SCARA robot is used. Through the D-H parameters, the inverse kinematics is calculated. The D-H parameters are shown in the Table 1 (Alshamasin, Ionescu, and Al-Kasasbeh 2009).

The result from inverse kinematics input to predefined job as angle of robot's joints and the path planning is done. The result of path planning is shown in Figure 10. After path planning is done, it is possible to change the position of

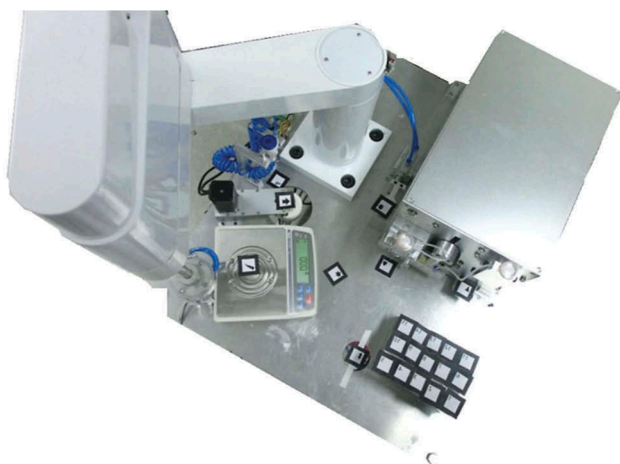


Figure 8. Marker-installed robotic system.



Figure 9. Marker extraction result.

devices and it doesn't matter to worry about a change of space. Also, it is possible to add or adjust process easy. As user, to exclude time to make predefined job, it takes minutes for just installing markers. However, with lead through method, it takes 6 h for a practised hand.

6. Conclusion

Path planning methods have previously been studied for solving the problem with robotic system. With the rising popularity of automation in many fields, the path planning comes important with increasing amount of amateurs' usage.

The objective of path planning by using marker detection with SCARA robot is to develop and implement for amateurs. The objective can be divided into three steps. One is extracting markers' position and direction data image from 3D vision sensor through image processing. Another is to complete developing process of the path planning with using extracted markers' physical data. The last is implementation to PET dispensing system with using SCARA robot.

First of all, the marker extraction process is setup. It is image processing that extract markers' pixel position and direction data from colour image. Especially, blob labelling and feature extraction is used for detecting markers. Using stereo camera, Kinect sensor and data of colour image replace with physical position and direction with depth image.

Second, the path planning method is defined with using detected markers' information. For users, the predefined script and marker images are made for the goal of system. It helps users to install robotic system without considering any role of

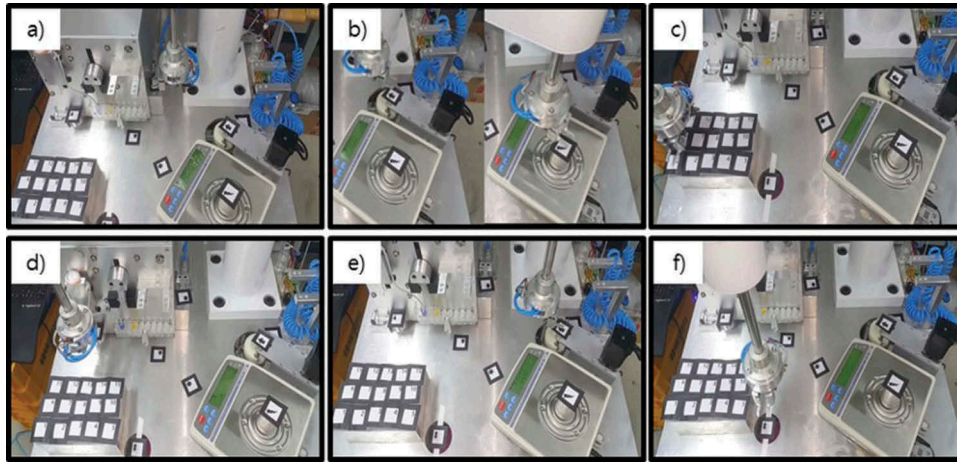


Figure 10. Path planning with markers for PET dispensing system.

Table 1. D-H parameters of the SCARA Robot.

Axis	Θ	D	a	α	Range
1	θ_1	0	$l_1 = 300 \text{ mm}$	0	$110^\circ \sim 110^\circ$
2	θ_2	0	$l_2 = 350 \text{ mm}$	0	$137^\circ \sim 137^\circ$
3	0	d_3	0	0	$0^\circ \sim 150^\circ$
4	θ_4	0	0	0	$180^\circ \sim 180^\circ$

each component. It cause that the role of experts get smaller and smaller in installing robotic system.

Finally, the path planning method is implemented to PET robotic dispensing system with using SCARA Robot. PET robotic dispensing system needs variable type of path planning depending on space. Also, the system can be installed and applied easy and fast by amateurs.

This article suggested SCARA robot's path planning method by using marker detection. This method used simple markers and a Kinect sensor. A non-robotic expert worker can deal with changing environments easily for any SCARA robot system. In its current method, it has a restricted application for SCARA robot. However, this method can be complementary to other industrial robots and other path planning methods. With collaborating methods or applying to another robotic systems, the method can be developed as path planning which help amateurs' path planning method.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This research was supported by Korea Institute for Advancement of Technology (KIAT), funded by the Ministry of Trade, Industry and Energy (MOTIE) under Grant [N0001594].

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